Ravindra Chowdary Jonnagadla

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Education

SRM Institute of Science and Technology Chennai, India
B.Tech in Computer Science and Engineering – Cloud Computing (CGPA: 8.01/10) May 2026

Technical Skills

Languages: Python, C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, Bash, Shell Scripting

AI/ML & Research: PyTorch, TensorFlow, Scikit-learn, Hugging Face, NumPy, Pandas, SciPy, WGAN-GP, Reinforcement Learning (PPO), NLP, Multimodal Learning, EEG Signal Processing

Generative AI & Speech: Whisper, Faster-Whisper, Sentence Transformers, ONNX Runtime, Model Quantization, FFmpeg, MoviePy, STT/TTS, Retrieval-Augmented Generation (RAG)

Backend & Systems: FastAPI, Flask, Django, Node.js, Next.js, React, RESTful APIs, WebSockets, Redis, BullMQ, Celery, Microservices, Browser Automation (Selenium), YouTube Data API

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM), Cloudflare R2, Docker, Kubernetes, Terraform, GitHub Actions, Linux, Git, Prometheus, Grafana

Data & Search: ChromaDB, BM25, Elasticsearch, Logstash, Kibana, ETL Pipelines, Vector Search

Certifications: Oracle Certified Professional: MySQL 8.0 Database Developer (1Z0-909) | AWS Certified Cloud Practitioner

Professional Experience

Waani – Conversational Voice AI Platform Chennai, India
Lead Engineer & Co-Founder July 2026 – Present

- Architected an ultra-low-latency duplex voice pipeline orchestrating telephony protocols, streaming STT (Sarvam AI), and LLM token inference, achieving $<$**800ms p95 voice-to-voice turn-around time** across 5,000+ synthetic concurrent voice sessions.
- Engineered conversational state machines and dynamic intent branching for automated lead qualification and appointment scheduling, configuring automated fallback logic for human operator handoff.
- Built an asynchronous dual-channel audio ring buffer with edge Voice Activity Detection (VAD) over WebSockets, implementing atomic cancellation tokens to flush in-flight TTS buffers on instant user barge-in.

LynkZen – Instagram DM Automation SaaS Chennai, India
Founding Engineer & Builder April 2026 – June 2026

- Built and launched an event-driven Instagram automation engine using Next.js, TypeScript, and Tailwind CSS, decoupling webhook ingest via asynchronous worker queues to absorb 10k+ burst events from comment-to-DM triggers and story replies.
- Developed an interactive node-based visual workflow builder allowing creators to configure multi-step conditional messaging funnels and custom automated response paths without writing code.
- Scaled platform to **3,000 active beta signups for closed testing**, implementing token-bucket rate limiting via atomic Redis Lua scripts to eliminate race conditions and guarantee zero 429 quota violations against Meta Graph API limits.

Peer-Reviewed Research & Technical Projects

AMERS: Multimodal Emotion Recognition System | *PyTorch, Signal Processing* **IEEE ICETEA 2026**

- Developed a multimodal emotion recognition architecture synchronizing multi-channel EEG (128Hz) and speech spectrograms via Cross-Modal Mutual Attention (CMMA) to capture inter-modal temporal-spatial dynamics.
- Integrated an Emotion-Aware Gating (EAG) WGAN-GP generator with a PPO reinforcement learning controller, dynamically scheduling synthetic-to-real training batches against discriminator reward stability.
- Evaluated on DEAP and SEED benchmarks under Leave-One-Subject-Out (LOSO) cross-validation across 4 emotion classes, achieving **82.55% accuracy** (+7.4% over unimodal baselines), **0.83 macro F1-score**, and **0.77 Cohen's kappa**.

SmartCap: Automated Subtitle & Transcoding Pipeline | *Python, Whisper, FFmpeg* **IEEE ICICV 2025** (Doc ID: 11085491)

- Built an automated transcription engine combining OpenAI Whisper with acoustic timestamp alignment to generate frame-accurate synchronized video subtitles (.srt/.vtt); published as Sole/Lead Author.
- Architected an asynchronous worker pipeline orchestrating FFmpeg stream filters for dynamic audio normalization and hardcoded subtitle burning, isolating CPU-bound transcoding to achieve a **0.35x Real-Time Factor (RTF)**.

Smart QR-Based University Parking System | *Flask, Python, PostgreSQL, Twilio API* **IEEE CONIT 2025** (Doc ID: 11166870)

- Designed a campus parking management application with HMAC-SHA256 signed dynamic QR codes, timestamped nonces cached in Redis to prevent replay attacks, and Twilio OTP verification; published as Co-Author.
- Engineered slot reservation concurrency using optimistic row-level locking with versioning in PostgreSQL, maintaining **zero double-booking anomalies and 99.8% transaction integrity** under simulated 1,000-user rush-hour contention.